

Athena ClusterMessage wire format

Three header words $\Rightarrow$ optional body $\Rightarrow$ raw bytes for Data

Transfer sequence

Header • always
3 words • MPI tag 0



Body • when needed
10 words • tag header[2]



Data bytes
len • tag + 16384

Fixed header • $3 \times \text{uint32_t}$

header[0]
message type

header[1]
source rank

header[2]
event index / body tag / 0

WorkerStatus body • 10 words

0	1	2	3	4	5	6	7	8	9
stat	made	done	skip	0	0	0	0	0	0

0 status • 1 created

2 finished • 3 skipped

4..9 are zero

DataDescr body • 10 words

0	1	2	3	4	5	6	7	8	9
ptr_hi	ptr_lo	len_hi	len_lo	log2 align	dest ID	evt_hi	evt_lo	req_hi	req_lo
ptr		len				event		request	

64-bit fields = $(\text{hi} \ll 32) + \text{lo}$

dest chooses the memory resource.

Allocate len bytes with alignment align.

ptr then becomes receiver-local.

Field colours

header	status	counter	address	length	memory
event ID	request ID	raw bytes	unused		